A Fixture Paper on Robust Handover Prediction

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Abstract

We study handover prediction under spoofed measurement reports. Our detector flags 92 percent of spoofed reports at a 3 percent false positive rate in simulation.

1 Introduction

Predictive handover in dense urban deployments is challenging because measurement reports arrive with delay and noise. Prior work has studied recurrent predictors [3], [4] and reinforcement learning approaches [5]-[7]. Predictive handover in dense urban deployments is challenging because measurement reports arrive with delay and noise. Prior work has studied recurrent predictors [3], [4] and reinforcement learning approaches [5]-[7]. Predictive handover in dense urban deployments is challenging because measurement reports arrive with delay and noise. Prior work has studied recurrent predictors [3], [4] and reinforcement learning approaches [5]-[7].

2 System Model

We consider a macro cell layer with small cells. Each user equipment reports RSRP and RSRQ every 40 ms. The serving base station runs a learned predictor that outputs the target cell. Predictive handover in dense urban deployments is challenging because measurement reports arrive with delay and noise. Prior work has studied recurrent predictors [3], [4] and reinforcement learning approaches [5]-[7]. Predictive handover in dense urban deployments is challenging because measurement reports arrive with delay and noise. Prior work has studied recurrent predictors [3], [4] and reinforcement learning approaches [5]-[7]. Predictive handover in dense urban deployments is challenging because measurement reports arrive with delay and noise. Prior work has studied recurrent predictors [3], [4] and reinforcement learning approaches [5]-[7].

Method	Acc	FPR
A	0.91	0.03
B	0.87	0.05

Table 1: Results

3 Results

The proposed detector reduced successful spoofing attacks by 35 percent compared with the undefended baseline. Adversarial training reduced clean prediction accuracy by 1.5 points. Predictive handover in dense urban deployments is challenging because measurement reports arrive with delay and noise. Prior work has studied recurrent predictors [3], [4] and reinforcement learning approaches [5]-[7]. Predictive handover in dense urban deployments is challenging because measurement reports arrive with delay and noise. Prior work has studied recurrent predictors [3], [4] and reinforcement learning approaches [5]-[7]. Predictive handover in dense urban deployments is challenging because measurement reports arrive with delay and noise. Prior work has studied recurrent predictors [3], [4] and reinforcement learning approaches [5]-[7].

3.1 Ablation

Removing the temporal consistency feature increased the false positive rate to 7 percent.

4 Conclusion

Detection of spoofed measurement reports is feasible with low overhead. Predictive handover in dense urban deployments is challenging because measurement reports arrive with delay and noise. Prior work has studied recurrent predictors [3], [4] and reinforcement learning approaches [5]-[7]. Predictive handover in dense urban deployments is challenging because measurement reports arrive with delay and noise. Prior work has studied recurrent predictors [3], [4] and reinforcement learning approaches [5]-[7]. Predictive handover in dense urban deployments is challenging because measurement reports arrive with delay and noise. Prior work has studied recurrent predictors [3], [4] and reinforcement learning approaches [5]-[7].

References

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